

Hari Sudarsan

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Work Experience

Founding Engineer — Nava AI

July 2025 – Present

SDE, Onsite — Bengaluru

- Built and maintained software-defined networking (SDN) layer using OVN/OVS for a multi-tenant cloud platform, including VLAN-based network isolation and policy enforcement
- Developed networking firewall constructs—security groups, ACLs, and packet filtering rules—using OVN, implementing layer 2–4 traffic control
- Researched and prototyped eBPF-based data plane acceleration for OVN/OVS, evaluating offload strategies for high-throughput packet processing
- Implemented SSH gateway for secure VM access, handling TCP connection multiplexing, authentication, and tunnelling
- Engineered service-to-service communication layer incorporating service mesh and east-west traffic routing patterns
- Developed network manager service (Golang) for region-level networking constructs, orchestrating VPCs, subnets, and floating IPs
- Authored Helm charts for automated, reproducible deployment of networking and compute infrastructure across regions

Software Engineering Intern — Accuknox

May 2025 – Jul 2025

SDE Intern, Remote

- Reduced KubeArmor CLI CPU usage by 25% by optimizing process and network event retrieval paths
- Enhanced BPF LSM filtering logic for fine-grained network and filesystem policy enforcement at the kernel level
- Implemented OpenSearch integration and Grafana datasource for network and process observability

Software Engineering Intern — Safedep

Jan 2025 – Mar 2025

SDE Intern, Remote

- Built malware analysis tool in Golang with multi-ecosystem package scanning (Go, npm), including network dependency graph analysis
- Designed CI/CD pipeline with automated testing and reporting for continuous security scanning

LFX Mentee — KubeArmor (CNCF)

Mar 2024 – Jul 2024

SDE Intern, Remote

- Contributed DNS visibility feature (kprobes-based) to upstream KubeArmor, enabling kernel-level DNS query monitoring for runtime security
- Submitted multiple upstream PRs for operator RBAC updates, hardening rule presets, and relay server integration
- Developed Grafana plugin (Golang/TypeScript) processing Elasticsearch logs to construct process and network topology graphs using Kubernetes informers for IP resolution
- Built KubeArmor OpenTelemetry adapter for integrating Elasticsearch into the observability pipeline

Education

Indian Institute of Technology (BHU), Varanasi

Aug 2025

B.Tech, Metallurgical Engineering

Open-Source Projects

Kronos

Runtime Enforcement Engine using BPF-LSM

- Built a network and security policy enforcement engine (Rust/Aya) leveraging BPF-LSM for real-time MAC policies on Kubernetes pods
- Restricts network access (including DNS filtering), filesystem operations, and Linux capabilities at the kernel level
- Implements CRD-based policy management using kube-rs — github.com/harisudarsan1/Kronos

DNS-tracer

eBPF-based DNS Query Capture

- Developed an eBPF program (C) tracing DNS queries at the syscall level using kprobes and tracepoints on UDP DNS traffic
- Hooks directly into kernel networking paths for low-overhead packet inspection

ebpf-profiler

Kernel-level Profiling via eBPF

- Built a profiling tool attaching eBPF programs to perf events for CPU, network I/O, and disk I/O metrics collection
- In-kernel data collection for lower overhead compared to traditional syscall-based monitoring

Open-Source Community

- Active contributor to **KubeArmor** (CNCF sandbox project) with merged upstream PRs for DNS visibility (kprobes), operator RBAC, and CLI performance optimization
- Published technical blog on networking and infrastructure topics — [blog repository](#)
- Maintainer of multiple open-source tools including Kronos (BPF-LSM enforcement) and DNS-tracer

Skills & Interests

- **Languages:** C, Golang, Rust, TypeScript
- **Networking:** OVN/OVS, eBPF, TCP/UDP, VLAN, CNI, DNS, DHCP, Firewall, ACLs, SDN, Service Mesh
- **Linux Kernel:** BPF-LSM, kprobes, tracepoints, TC hooks, perf events, Networking Stack
- **Infrastructure:** Kubernetes, Docker, Helm, KubeVirt, OpenShift, CI/CD, OpenTelemetry
- **Interests:** Linux Kernel Networking, eBPF, Software-Defined Networking, Open Source, Cloud Infrastructure